

Feasibility study and energy conversion analysis of stand-alone hybrid renewable energy system



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ABSTRACT

There is a great interest in the development of renewable power technologies in Algeria, and more particularly hybrid concept. The present paper has investigated the performance of hybrid PV–Wind–Diesel–Battery configuration based on hourly measurements of Adrar climate (southern Algeria). Data of global solar radiation, ambient temperature and wind speed for a period of one year have been used. Firstly, the proposed hybrid system has been optimized by means of HOMER software. The optimization process has been carried out taking into account renewable resources potential and energy demand; while maximizing renewable electricity use and fuel saving are the purpose. In the second step, a mathematical model has been developed to ensure efficient energy management on the basis of various operation strategies. The analysis has shown that renewable energy system (PV–Wind) is able to supply about 70% of the demand. Wind power has ranked first with 43% of the annual total electricity production followed by diesel generator (with 31%) while the remaining fraction is being to PV panels. In this context, 69% of the fossil fuel can be saved when using the proposed hybrid configuration instead of the diesel generators that are currently installed in most remote regions in Algeria. Such a concept is very promising to meet the focus of renewable energy program announced in 2011.

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1. Introduction

According to the International Energy Agency (IEA) the global energy consumption will increase by 37% to 2040 [1]. Furthermore, it has been expected a significant raise in the world electricity demand by 80% over the period 2012–2040 [1]. As a result, fossil fuel prices are projected to move forward in the next 20 years to reach \$215/barrel in 2035 [2]. Nowadays, current energy policies are focusing on the concept of energy efficiency gains and low carbon fuels [1]. The penetration of low-carbon technologies in recent years and the enhancements in the efficiency of energy conversion systems have just slowdown the demand growth. It is worthy to note that the energy prices are strongly related to the turmoil in parts of the Middle East and North Africa (MENA). Therefore, the fluctuations in fossil fuels prices in the last decade together with the coming shortage and depletion have heighten concerns over future energy supply security. To this end, advanced energy policies, in many countries, have been based

on the development of renewable and sustainable energy resources [3–7].

Following this approach, Algeria has announced, in March 2011, a very promising program to develop and expand the use of renewable energies in order to move from the fossil fuels era to the green energy era by 2030 [8]. Algeria is aimed through this program to be a pioneer in the production of electricity from solar photovoltaic and concentrating solar thermal power (CSP). Therefore, solar energy is the major focus of the Algerian renewable energy program and it is expected to supply about 37% of the national electricity need by 2030 [8].

Day than day, the use of both CSP and photovoltaic technologies around the world, particularly in Spain, Germany and the US, has shown a significant cost reduction through efficiency improvements and scaling up [9,10]. Whereas CSP options are more preferred at large scale applications, the photovoltaic has proven their performance at small scale and off-grid use in remote regions. The competitiveness at small scale applications, easy maintenance and installation, and lower energy cost in regions where electrical grid is not available are the most important advantages of solar photovoltaic technology [11].

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Nomenclature

A_c	PV panels receiving area	T_c	cell temperature
C_{bat}	capacity of the battery at time (t)	V_c	cut-off wind speed
C_{batmax}	maximum capacity of the battery	V_d	cut-in wind speed
DOD	deep of discharge	$V(H)$	wind speed measured at the hub height
E_s	global solar irradiation	V_n	rated wind speed
N_{mod}	number of PV modules		
P_{cha}	power demand	Greek symbols	
P_f	full module factor	α	wind speed power law coefficient
P_{ge}	power produced by the diesel generator	η_{gen}	efficiency of PV generator
P_n	nominal power	η_{mod}	efficiency of the PV module
P_{pvs}, P_{ws}	electricity produced by PV and wind generators respectively	η_{ref}	efficiency reference of PV generator
P_w	power produced by the wind turbine	$\eta_{dc/ac}$	converter efficiency
		η_{decha}	efficiency of battery discharging process

For the case of Algeria this latter advantage is of a particular interest because of its large area and lower density of population. For instance, the population density in the part of Sahara is less than one people per one square kilometer. Covering such a wide region with power grid is an expensive investment. Even, if it were be, the cost of electricity will be higher than actual average values. In such a context, photovoltaic panels would be the best solution for both population and government. In order to do so, the ministry of energy and mines has planned to install several solar photovoltaic projects with a total capacity of 800 MW by 2020 as a part of the program highlighted above. Moreover, other projects with an annual capacity of 200 MW are planned between 2021 and 2030 [8]. At this year, the electricity from PV should be competitive to the state-of-the-art power generation options. Recent studies have indicated that about a half in the cost reduction can be achieved though scaling up while the other half is the concern of R&D activities [9].

To this end, most recent R&D activities worldwide are focusing on the combination of solar photovoltaic with other sources of energy as it is the only alternative to overcome the issue of higher electricity cost up to now. Many countries are now showing an increasing interest in the development of hybrid solar PV power systems. Emerging Markets such as China, India, Indonesia, Tunisia, Algeria, and Saudi Arabia are excellent examples. For China, Li et al. [12] have investigated the feasibility of a hybrid PV–Wind with storage system to electrify a household in the region of Urumqi. When compared with only PV and only Wind power plants, the proposed configuration has shown a reduction in the total net present cost by about 9% and 11% respectively. Rehman et al. [13] have interested on the feasibility of a modifying a diesel generator into hybrid PV–wind–diesel plant in Rafha, Saudi Arabia. They have found that at a diesel price of 0.6 \$/L the hybrid system become cost competitive. With regard to the efficiency and system reliability, Daud and Ismail [14] have designed and analyzed a PV–wind–diesel hybrid system for a family house in Palestine. Taking into account the actual energy resources map in Tunisia, Colantoni et al. [15] have developed a mathematical program to find the optimum dimension of hybrid PV–Wind power system. The authors have highlighted the potential of hybrid fossil renewable power plants to reduce fossil fuels import in Tunisia. Celik [16] have introduced a novel sizing methodology for hybrid Wind–PV system based on the monthly variation in the required size of the renewable energy converters. Kumar and Manoharan [17] have dealt with the feasibility of implanting hybrid PV–Diesel plant in Tamil Nadu, India. The economic analysis based on the net present cost, fuel consumption and renewable electricity fraction have indicated that the interior climatic zone is the

most suitable for hybrid photovoltaic–wind power generation plants. Hafez and Bhattacharya [18] have analyzed the potential of installing hybrid PV–wind–diesel system in Johor Bahru, Malaysia. Akms et al. [19] and Hoque et al. [20] have been interested on the PV–wind–diesel generator hybrid power system under Bangladesh climates. The former investigated the performance of the hybrid system under Martin Island climatic conditions whereas the latter has considered a small village in Comilla. Lal and Raturi [21] have carried out an economic assessment of PV–Wind–Diesel hybrid power system for a remote location on the island of Vanua Levu.

In Algeria there is also an increasing interest on the development and investigation of hybrid photovoltaic systems, particularly after the introduction of the renewable energy program in 2011. Khelif et al. [22] have proposed the combination of PV panels with an existed diesel generator under operation in AFRA (southern Algeria). The authors have developed a mathematical program to predict the hourly performance of the proposed hybrid plant including voltage, current, fuel consumption, battery state of charge and electricity production. The model is also capable to perform an economic assessment to evaluate the profitability of the power system. They have concluded that the profitability of the proposed hybrid system is strongly related to fossil fuel cost and beyond a 0.34 €/l of fuel the project is cost-effective. Rezzouk and Mellit [23] have interested in the feasibility of powering a research unit (UDES) located in the north of Algeria with a hybrid PV–Diesel–battery system. HOMER program has been applied and various configurations as a function of PV penetration including a stand-alone diesel generator and stand-alone PV system were considered. The authors have found that a hybrid power plant with 25% electricity production from PV panels is the optimum configuration that ensures high stability with the lowest electricity cost. Himri et al. [24] have proposed the combination of an existing diesel generator with wind turbine to feed a remote region in the southern of Algeria. They have used HOMER program to evaluate the energy production, life-cycle costs and greenhouse gas emissions reduction of the proposed Wind–Diesel power system. The simulation has shown that the hybrid system becomes feasible at a wind speed of 5.48 m/s with a fossil fuel price of 0.162 \$/L or more. Aissou et al. [25] have experimentally and numerically studied the performance of small scale PV–Wind power system under coastal climate in Bejaia, Algeria. The hybrid plant consists of six photovoltaic panels of 175 W each connected in parallel, 1 kW wind turbine, batteries, inverter, and sensors for measurements. The hybrid plant control and power management have been evaluated. The design and modeling of the components have been carried out using Matlab/Simulink while the power control strategy is

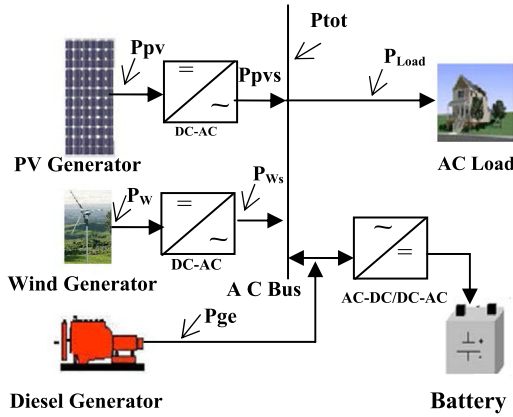


Fig. 1. Schematic block of the proposed hybrid system.

Table 1

Technical specifications of wind turbine Proven 15.

Technical data	Value	Unit
Nominal power	15	kW
Nominal wind speed	12	m/s
Minimum wind speed	2.5	m/s
Maximum wind speed	70	m/s
Weight of rotor	1200	kg
Rotor diameter	9	m
Height from the ground	25	m
Nominal voltage	300	V
Lifetime	25	Years

implanted though LabVIEW software. The authors have experimentally demonstrated the potential of the proposed hybrid system as well as its management control strategy.

The integration of photovoltaic with both wind power and fossil fuels has been also the subject of many published papers. In Algeria, the combination of various sources of energy (renewable and fossil) can offer a significant cost reduction of power generation, provides continuous and stable power, reduces O&M costs and saving fossil fuels for future generations. To highlight this fact, Zeraïa et al. [26] have analyzed numerous hybrid configurations of PV–Wind–Diesel with storage to determine the optimum concept for Algerian remote regions. The performance prediction of the selected configurations have been carried out under six different climates, i.e., Algiers, Djelfa, Ghardaïa, Adrar, Tindouf and Tamanrasset. The simulation using HOMER program has pointed out that hybrid PV–Diesel power system is the most suitable. In order to determine the optimum size of stand-alone hybrid PV–Wind–Diesel–Batteryplant, Saheb-Koussa et al. [27] have developed a model in Matlab/Simulink. Various locations in Algeria with different climates have been selected, i.e., Bouzareah (coastal), Batna and El Oued (high plateaus), Ghardaïa, Adrar and Tamanrasset (hot dry climates). The authors have found that the optimum configuration in each site is strongly related to their renewable resources including solar radiation intensity and wind potential.

In this paper we have investigated the feasibility of hybrid PV–Wind–Diesel–Battery power system under the southern Algerian climate whereby the performances of each part have been analyzed. To do so, weather data including ambient temperature, solar radiation intensity and wind speed have been used and a predetermined load profile over a week was applied for testing. After the optimization of the size of components using HOMER program, the energy management of the power plant has been carried out to maximize renewable energies utilization and minimize fossil fuel consumption. This latter step has been achieved through a

model developed using Matlab software. In the next section the considered hybrid system is described. The modeling and technical specifications of each component are detailed in Section 3. Section 4 illustrates site data while Section 5 concerned with the optimization of the hybrid system. The developed model for energy management is highlighted in Section 6 and the results of the present analysis are discussed in Section 7.

2. The proposed hybrid system

The proposed hybrid plant consists of a photovoltaic panels, Wind turbine, diesel generator, battery bank for storage, and inverters. Fig. 1 illustrates the design of the hybrid system. The inverter converts the electricity into either a DC or AC current depending on the direction of power. The battery requires a DC current to charge while the load requires an AC current. The output of the PV panels and wind generator is DC whereas the output of the diesel generator is AC. That is why an inverter has been included for optimum power control and conversion.

In the operation strategy solar PV and Wind Turbine are combined together to cover the primary load and any excess in the electricity will be stored in the battery. In the case when renewables (PV and Wind) are not enough to meet the demand, the battery is used to supply the shortage. When there is insufficient energy from either the battery or PV–Wind system, the diesel generator will automatically start to bring back the complement and to recharge the battery through a bidirectional inverter.

The operation strategy proposed in this study is aimed at:

- Minimizing fossil fuel consumption.
- Maximizing renewable energies utilization (solar PV and Wind).
- Minimize the storage capacity of the battery.
- Reducing the operation duration of the diesel generator and therefore enhances its reliability, reduces O&M costs and GHG emissions.

3. Design specifications and modeling of the hybrid system

3.1. Wind turbine

The wind turbine Proven 15 has been selected in this study. Rated power of the turbine is 15 kW at wind speed 12 m/s. In Table 1, it is highlighted the design specifications of this component. Minimum, maximum wind speed for operating the wind turbine, its lifetime and other parameters are given.

The power curve in Fig. 2 describes the amount of power produced by the turbine as function of wind speed at hub height. The power output of the wind turbine is strongly related to wind speed. Various models have been applied calculate the energy produced by the wind turbine [28,29]. The quadratic model, illustrated in Eq. (1) is selected in this paper.

$$P_w(V_v) = \begin{cases} P_n \cdot \frac{V_v^3 - V_d^3}{V_n^3 - V_d^3} & V_d < V_v < V_n \\ P_n & V_n \leq V_v < V_c \\ 0 & V_v \leq V_d \text{ et } V_v \geq V_c \end{cases} \quad (1)$$

where P_w is the power produced by the wind turbine, P_n is the nominal power, V_d is the cut-in wind speed, V_n is the rated wind speed and V_c is the cut-off wind speed.

The adjustment of the wind profile for height is taken into account by using the power law which is a useful tool to model the vertical profile of wind speed. The model which is illustrated in Eq. (2) is used for wind speed estimation [30].

$$V(H) = V(H_m) \left(\frac{H}{H_m} \right)^\alpha \quad (2)$$

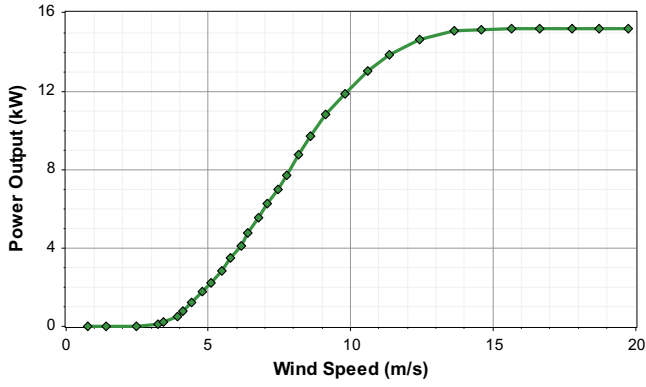


Fig. 2. Power curve of the wind turbine given by the manufacturer.

where $V(H)$ is the wind speed measured at the hub height H_m , $V(H_m)$ is the wind speed measured at the reference height H_m and α is the wind speed power law coefficient.

3.2. Photovoltaic modules

The PV module is made up of photovoltaic cells and it is used to convert solar radiation into electricity. The PV module (Type: KD 180 GX-LP) from KYOCERA has been chosen in the study. The basic information of the selected PV module is summarized in Table 2.

The energy output of the PV generator is calculated based on the following equation:

$$P_{gen} = \eta_{gen} \times A_c \times E_s \quad (3)$$

A_c is the PV panels receiving area, η_{gen} is the efficiency of PV generator and E_s is the global solar irradiation.

The efficiency of the PV generator is a function of ambient conditions and PV cell proprieties [31]:

$$\eta_{gen} = \eta_{mod} \times \eta_{dc/ac} \times P_f \times N_{mod} \quad (4)$$

In Eq. (4) η_{mod} is the efficiency of the PV module, $\eta_{dc/ac}$ is the electrical efficiency of the DC/AC inverter, P_f is the full module factor and N_{mod} is the number of PV modules.

The efficiency of the photovoltaic module can be estimated by the following equation:

$$\eta_{mod} = \eta_{ref} \{1 - \beta(T_c - T_{ref})\} \quad (5)$$

Table 2
Technical data of the PV module.

Technical data	Value	Unit
<i>Performance under ISO conditions</i>		
Maximum power	180	W
Maximum voltage	23.6	V
Maximum current	7.63	A
Open circuit voltage (Voc)	29.5	V
Shortage current (Isc)	8.35	A
Maximum voltage of the system	600	V
Temperature coefficient for Voc	-1.06×10^{-1}	V/°C
Temperature coefficient for Isc	5.01×10^{-3}	A/°C
<i>Performance under 800 W/m² irradiance</i>		
Maximum power	127	W
Maximum voltage	20.8	V
Maximum current	6.1	A
Open circuit voltage (Voc)	26.5	V
Shortage current (Isc)	6.8	A
<i>PV module features</i>		
Number of cell per module	48	cell
Dimension (length × width × thickness)	1341 × 990 × 36	mm
Weight	16.5	kg

With β is a coefficient accounting the PV module efficiency as function of temperature, η_{ref} is the efficiency reference of PV generator, T_c is the Cell temperature, $\eta_{dc/ac}$ is the converter efficiency equals to 0.95 and P_f is the full module factor equals to 0.9.

The cell temperature can be calculated by:

$$T_c = T_{amb} + E_s \left(\frac{NOCT - 20}{800} \right) \quad (6)$$

With T_{amb} is ambient temperature, E_s is the global solar irradiation at the selected site, NOCT is the operation temperature of PV cell under preselected conditions (global radiation of 800 W/m², ambient temperature equals to 20 °C et mass air AM supposed to be equals to 1).

Eq. (7) provides the total electric power produced by both PV and Wind generators.

$$P_{tot}(t) = P_{pvs}(t) + P_{Ws}(t) \quad (7)$$

With $P_{pvs}(t)$ and $P_{Ws}(t)$ are the electricity produced by PV and wind generators respectively after passing through the inverter.

3.3. Battery bank

The battery bank is used to serve the required load. If the battery bank is less than minimum level, the diesel generator starts supplying the demand. The capital cost of batteries is \$3680 while the annual O&M cost is 180 \$/kW h [32]. Table 3 provides more details about the technical specification of the battery bank.

The following parameters are used for characterizing the battery:

- Nominal capacity Batman.
- The state of charge SOC.

$$SOC(t) = C_{bat}(t)/C_{bat \max}(t) \quad (8)$$

$(0 \leq SOC \leq 1)$

With C_{bat} is the capacity of the battery at time (t), $C_{bat \max}$ is the maximum capacity of the battery. If $SOC = 1$, the battery is full; If $SOC = 0$, the battery is empty.

The Charge and discharge processes can be evaluated by Eqs. (9) and (10) respectively.

$$C_{bat}(t) = C_{bat}(t-1) + ((P_{tot}(t) - P_{cha}(t)) + P_{ge}(t))\eta_{ac/dc}\eta_{cha}\Delta t \quad (9)$$

$$C_{bat}(t) = C_{bat}(t-1) + (P_{tot}(t) - P_{cha}(t))\Delta t/\eta_{dc/ac}\eta_{decha} \quad (10)$$

where P_{cha} is the power demand, P_{ge} is the power produced by the diesel generator, η_{decha} is the efficiency of battery discharging.

To protect the battery from overcharge and deep discharge the accumulator is used to control the state of charge according to the following equation:

$$C_{bat \min}(t) \leq C_{bat}(t) \leq C_{bat \max}(t) \quad (11)$$

The minimum capacity ($C_{bat \min}$) of the battery is given by:

$$C_{bat \min} = DOD \cdot C_{bat n} \quad (12)$$

With DOD is the deep of discharge; it is supposed to be equal to 0.5.

Table 3
Design features of the battery.

Technical data	Value	Unit
Voltage	48	V
Discharge current	192	A
Power	9.2	kW
Nominal capacity	1923	Ah
Minimum allowable load	50	%
Battery cost	3680	\$
O&M costs	180	\$/kW h
Lifetime	10	Years

3.4. Diesel generator

The diesel generator 404C-22G of PERKINS manufacturer has been selected in this analysis. Its maximum capacity is about 18.5 kW. The technical data of this generator are given in Table 4.

A linear model has been assumed to predict the performance of the diesel generator at partial load. Table 5 illustrates the variation of fuel consumption as function of power output from the selected generator.

3.5. Converter

Any system that contains both AC and DC elements requires a converter. In the proposed hybrid system the converter is applied for two purposes, i.e., to convert AC power from diesel generator to DC power used to charge the batteries. In this case it is called converter. It can also work as an inverter (DC–AC) to serve AC loads.

A power converter is required for a hybrid wind/PV/battery power system to maintain the flow of energy between DC and AC components.

4. Site and weather data description

4.1. Site features and load profile

In this analysis we have considered the site of Adrar province, southern Algeria (Latitude: 27.82°N, Longitude: 0.18°W, Altitude: 263.8 m). The climate of the site is tropical of the hot zone where the air is dry. Its features are significant differences between day and night time temperatures and a little uneven rainfall. The winter is short and cold and summer is long and hot. In Adrar there are important amount of fossil fuels recently discovered, moreover, a larger refinery for diesel fuel and many other hydrocarbons is now underway. The renewable potential including solar energy and wind are really abundant. These factors make the hybrid fossil-renewable power system very attractive in this region.

The daily typical load profile of the site village is highlighted in Fig. 3. It has been observed maximum loads of 125–130 kW between 8 and 10 p.m. because the site considered is inhabited area where most people are at home in this period. However, in the early morning most people are out of the homes and this significantly reducing electricity demand.

4.2. Solar radiation

The solar radiation data has been taken from a meteorological station of Adrar. The global solar irradiance recorded during a year (hour per hour) is shown in Fig. 4. It can be seen that the solar radiation is very important in this location. Higher radiation intensity has been observed from March to September while smaller amounts are available from October to February.

Table 4
Main design data of the diesel generator.

Technical data	Value	Unit
Constructor	PERKINS	
Type	404C-22G	
Rotation speed	1500	rot/min
Net power	18.5	kW
Cooling fluid	Water	
Electric current (DC)	12	V

Table 5
Fuel consumption as function of engine output.

Output power (kW)	Fuel consumption (L/h)
18.500	5.400
13.875	4.000
9.250	2.900
4.625	2.270

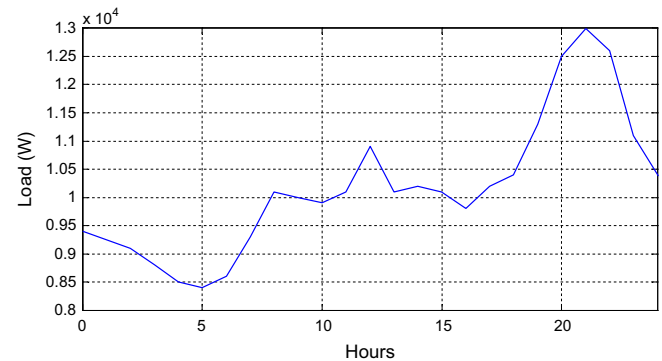


Fig. 3. Energy demand during a typical day.

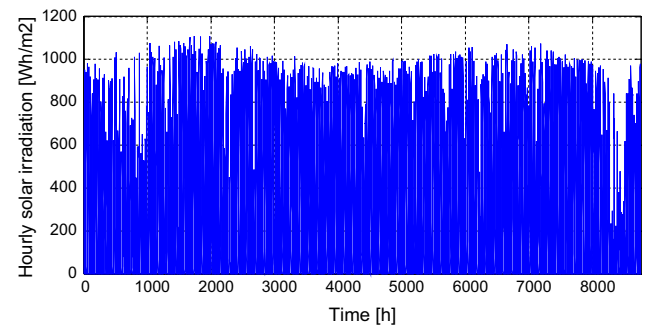


Fig. 4. Global radiation measured in Adrar during a period of one year.

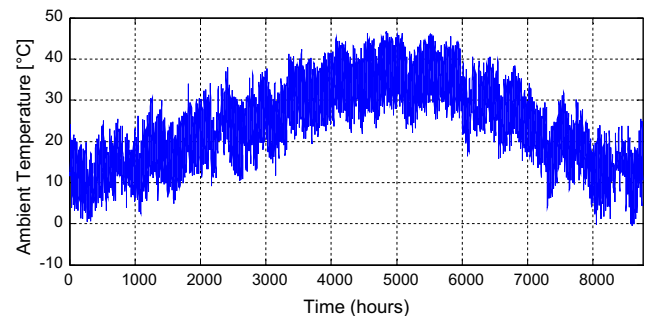


Fig. 5. Ambient temperature recorded in Adrar during a year.

4.3. Ambient temperature

The ambient temperature has a strong effect on the performance of PV panels and the density of air. Therefore, it is of a particular interest to have a measured data. The recorded ambient temperature during a year is illustrated in Fig. 5. It has been observed higher values during summer days; an average values higher than 40 °C have been recorded from June to August while from October to February the average ambient temperatures were around 23 °C. It is worthy to note that the annual average ambient temperature is about 36 °C.

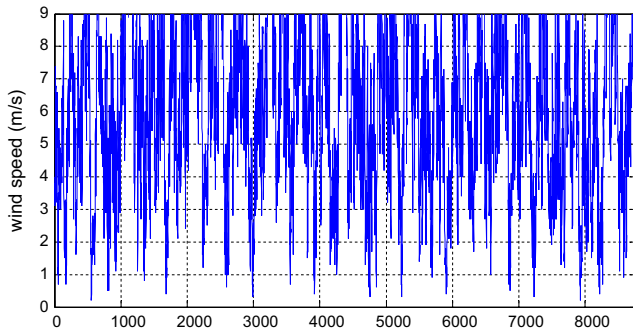


Fig. 6. Wind speed at the selected location during a year.

4.4. Wind speed

The wind speed data over a period of one year at a height of 10 m above ground level is shown in Fig. 6. At the selected site the wind speed ranges from 2 m/s to 9 m/s, and the annual average value is higher than 6 m/s. Higher average values have been recorded from September to March.

5. Optimization of the hybrid power system

5.1. About HOMER software

HOMER is stand for Hybrid Optimization Model for Electric Renewable. This software has been developed by the National Renewable Energy Laboratory (NREL) in 1993. It is one of the most widely useful tools in the simulation, optimization and sensibility analysis of renewable electricity systems in both off-grid and grid-connected concepts. HOMER offers a large number of components including wind turbines, PV panels, diesel generator, batteries, fuel cells, etc. For any hybrid power system the software is able to calculate the net present cost (NPC), renewable electricity fraction, greenhouse gases emissions and fuel consumption. This software simulates various hybrid renewable energy power systems and optimizes the size of their components on the basis of the NPC. Taking into account the load profile, renewable energy potential (solar, wind...), ambient conditions and the technical data of

components, the most cost effective configuration of the hybrid renewable energy system can be obtained using HOMER.

5.2. Optimizing the components size

In this study we have applied HOMER software for optimizing the hybrid PV–Wind–Diesel–Battery system. Numerous system configurations have been simulated to find the optimum one. To do so, the software requires the mean wind speed, global solar radiation and energy demand details, components technical specifications. All these data have been described in details in the previous sections. Comparisons of various designs of the hybrid systems regarding the technical and economic features of system components have been carried out to find the configuration has offer the lowest energy cost. The optimization process is mainly based on the net present cost. Note that the NPC is the total cost of installing and operating the hybrid system over its lifetime. This includes the initial cost, component replacements, operations and maintenance and fuel.

6. Energy management & operation strategy

The hybrid system is optimized to meet a typical energy demand. In order to maximize the renewable energy use and fuel saving, a mathematical model has been developed using Matlab/simulink. Matlab/simulink software is the most popular scientific computing system and array programming language, enables developers to perform complex mathematical computations from relatively simple instructions.

The flowchart of the model is illustrated in Fig. 7. The input data for the model are the hourly global irradiation, wind speed, ambient temperature, load profile and the technical data of the optimum configuration. Note that the optimum hybrid system is obtained by HOMER software. For each hour the model calculates the total available renewable electricity and then compares to the energy demand. When renewable electricity is sufficient to meet customers need, any excess in the electricity will be stored in the battery. This is the first preferred operation strategy. The second operation strategy is applied when renewable electricity is not enough to meet the load profile. In this case the model checks if the battery is able to supply the shortage. If this opportunity is not achievable, the diesel generator is used to supply the

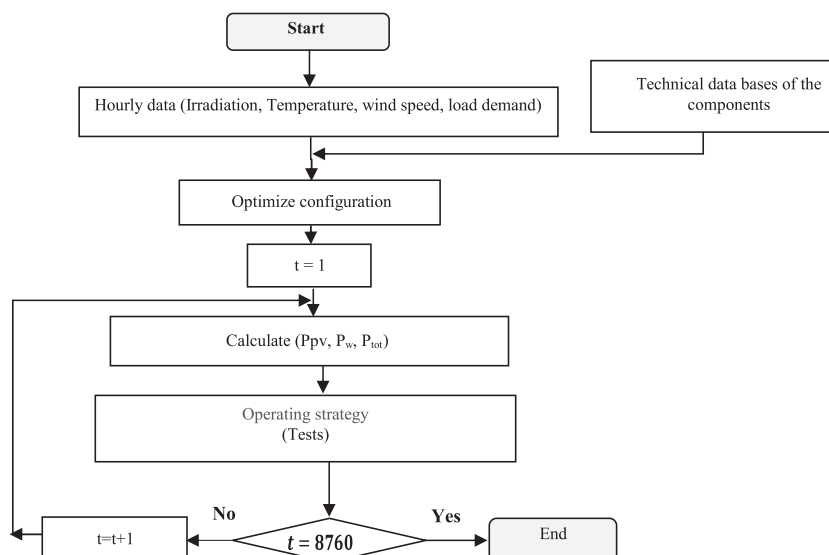


Fig. 7. Flowchart of the developed model for optimum energy management.

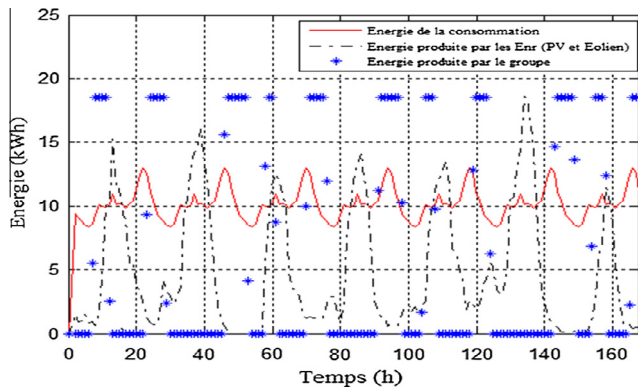
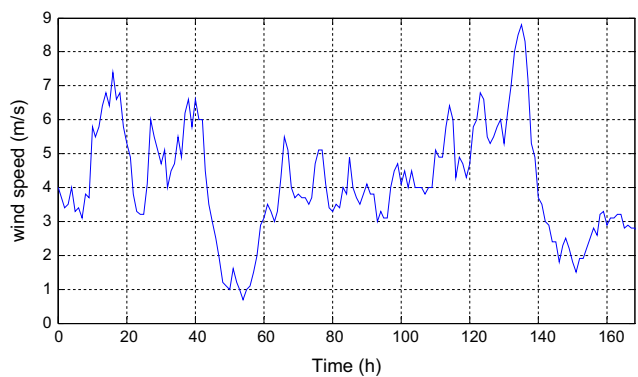
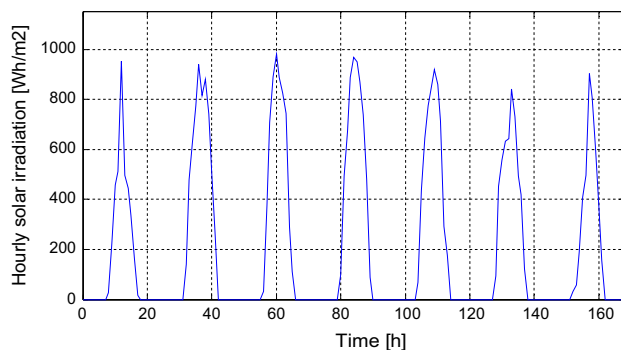


Fig. 8. The performance of the hybrid configuration during a period of one week.



(a) Wind speed



(b) Solar radiation

Fig. 9. Wind speed and solar radiation data over a week.

complement and to recharge the battery. This is the third operation strategy and it is the last solution since the proposed energy management is aimed to minimize the fossil fuel consumption and therefore reducing greenhouse gases emissions.

7. Discussion of the results

In the first step, the hybrid system has been analyzed during a period of one week. Fig. 8 shows the electricity production by the renewable system (PV + Wind) and diesel generator as well as energy demand. In Fig. 9 it is highlighted the solar radiation intensity and wind speed for better understanding the operation and management of the hybrid power plant.

It has been observed that renewable energy fraction is strongly related to solar radiation intensity, i.e., during sunny periods the

curve of renewable electricity is similar to the curve of the solar radiation. When the wind speed is below 7 m/s the fraction of energy from wind turbine is relatively lower. During the period from 130 h to 140 h (see Fig. 9a), the wind speed is higher than 7 m/s. This increase the energy produced by renewable system to reach 18 kWh. It is noticeable (from Figs. 8 and 9b) that energy consumption is higher than renewable electricity at night while it is lower during daylight when solar radiation intensity is important. The diesel generator works just few hours during night to meet the demand. The peak energy demand of about 13 kWh is occurred at night.

The more in-depth analysis has shown the following:

- At daylight (from 10 a.m. to 4 p.m.) the first operation strategy is applied since renewable system offers about 15 kW. During this time the energy consumption is lower than the renewable electricity, the remaining amount is stored in the battery.
- From sunset to 8 p.m. the second operation strategy is more preferred: during this time renewable electricity is not sufficient to meet the increased load and therefore the battery is used to supply the shortage.
- From 8 p.m. to sunrise: due to the increased demand the battery is discharging until it reaches its minimum. On the other hand, renewable is not enough to feed customer due to the absence of solar rays. To overcome this issue the third operation strategy is applied; the generator is used to supply the costumers and to recharge the battery.

For better understanding the operation of the proposed hybrid system, the annual performance has been investigated. The results are illustrated in Fig. 10. It can be seen that during sunny periods the renewable system (dashed curve) provides more energy than the customer need, while at night the increased demand is usually satisfied using the diesel generator. The annual sensitivity analysis has also indicated that wind power at Adrar is higher in autumn and winter (from 1 h to 3000 h and from 7000 to 8700 respectively). During daylight, even though solar radiation is higher in summer we have observed higher renewable electricity during winter and autumn (around 27 kWh) compared to summer (around 24 kWh). This increase is due to the higher wind speed during these seasons.

An important finding is that the decrease in solar electricity during winter and autumn periods is compensated by wind power which provides stable operation of the hybrid PV–Wind–Diesel–B battery system. This benefit is of a particular interest and makes the region of Adrar very promising to hybrid PV–Wind systems.

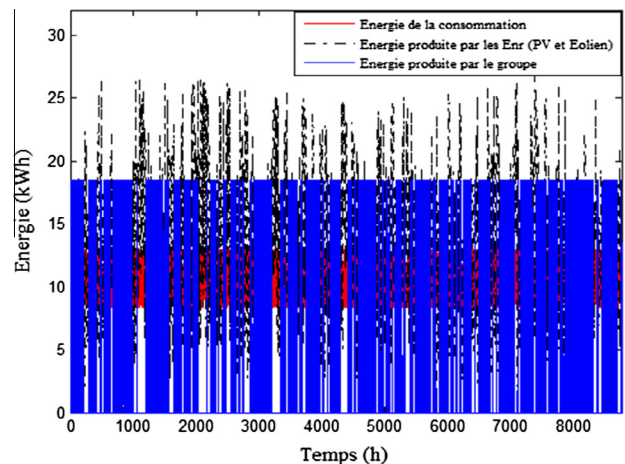


Fig. 10. Annual performance of the hybrid system.

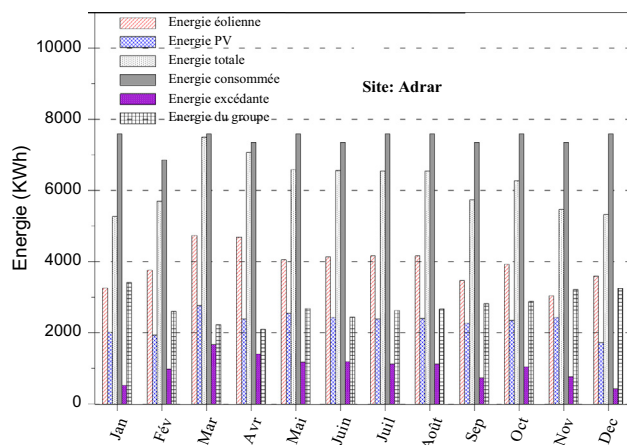
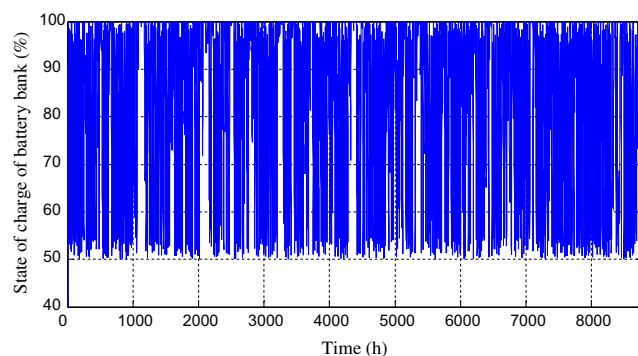
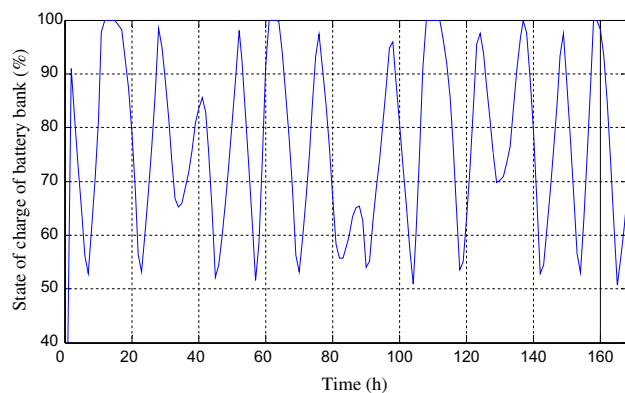


Fig. 11. Monthly energy production by each part of the hybrid system vs. energy consumption.



(a) state of charge of battery during a year



(b) state of charge of battery during a week

Fig. 12. Battery state of charge.

The monthly energy production of each component is shown in Fig. 11. A uniform load of about 7500 kW h per month is observed in the region of Adrar. The major fraction of this amount (about 6000 kW h) is provided by the renewable system including PV and wind. The average fraction of electricity from each component of the hybrid plant is highlighted in Table 6. The wind turbine is expected to provide about the half of the total power demand (43.69%) while the PV ranks second by (25.71%). The Diesel generator only used to meet the shortage during night. That is why its electricity fraction does not exceed 31%. This means that the hybrid system is saving about 70% of the fossil fuel consumption. The same fraction is the reduction in greenhouse gases emissions.

Table 6

Average annual electricity production by each component.

Components	Electric power (kW/year)	Fraction* (%)
Wind	46,922	43.69
PV panels	27,618	25.71
Diesel	32,868	30.60
Total	107,408	100

* Fraction = power produced from each components/the total energy produced.

The battery bank is also analyzed with awareness. In Fig. 12 it is represented the state of charge of the battery. Fig. 12a illustrates the battery load during year. The state of charge of the battery though a week is given in Fig. 12b. It have been observed that the charge–discharge cycle is relatively minor and this increase the lifetime of the battery.

8. Conclusion

A hybrid renewables-fossil power system has been investigated in this study whereby three types of energy are combined (solar photovoltaic, wind power and diesel fuel). A detailed analysis including hybrid system modeling, energy management and sensitivity study during short (week) and long (year) periods has been carried out. The hybrid configuration has been well optimized to ensure higher fuel saving while maximizing renewable electricity use is the target. The energy management introduced in this study was very efficient and allows rational energy balance between the three subsystems. It has been obtained that, under Adrar climate, the hybrid PV–Wind–Diesel–battery concept is capable to meet the customers need in remote regions. A fraction of about 43% of the total electricity demand has been supplied by the wind turbine while the PV panels provides around 26%. This is due to the higher solar radiation intensity and the wind speed though a year. Another important finding of the present study is that the proposed configuration can reduce the fossil fuel consumption by about 70% compared with only diesel generator that are usually installed to supply a remote regions in the southern of Algeria.

It is noteworthy that the actual diesel generators in operation are suffering from lower efficiency associated with serious fuel shortage which significantly increase the cost of electricity. Moreover, it slowdown the economic development of the remote regions and undoubtedly raise the social problems. In this context, hybrid renewables-fossil power plants might become the technology of choice in the coming decades as the shortage and depletion in fossil fuels together with the hike in their prices take a hold. If the Algerian renewable energy program is well implemented, such a hybrid system will be a lever for economic and social development, particularly through the establishment of wealth and job-creating industries.

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